Optimizing Deployment Configurations for LLM Inference: Challenges and Insights

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Abstract

Meta's Large Language Models (LLMs)—the *Llama* model family—serve nearly one billion monthly active users. Deploying these models for inference involves navigating a complex design space that spans diverse hardware options (e.g., H100, H200, MI300X), multiple parallelism strategies (tensor, pipeline, expert, context, and data parallelism), and nuanced runtime choices (e.g., continuous batching versus prefill-decode disaggregation)—all while leveraging workload-specific characteristics and meeting stringent service level objectives (SLOs). This paper presents insights we gained from developing and applying a systematic approach to analyze millions of deployment configurations and identify those that maximize throughput while meeting latency SLOs. We share lessons learned from our experience operating Llama inference at scale, including trade-offs among runtime designs, the phase-specific nature of parallelism strategies, opportunities for leveraging hardware heterogeneity, platform scaling behaviors, and system-level implications of model architectures such as Mixture-of-Experts (MoE). We hope our production experience offers practical insights for the broader LLM inference community.

1 Introduction

Large Language Models (LLMs), such as Llama (Touvron et al., 2023; Dubey et al., 2024), GPTs (Radford et al., 2018; 2019; Brown et al., 2020; Achiam et al., 2023), and Gemini (Comanici et al., 2025), have demonstrated exceptional capabilities across diverse tasks, transforming natural language processing (Yang et al., 2023; Wei et al., 2022; Chen et al., 2021; Liu et al., 2021; Zhang et al., 2019). Multi-modal models (Alayrac et al., 2022; Achiam et al., 2023) extend these capabilities to understanding and generating content across audio, images, and video.

With the availability of powerful open-source LLMs (Yan, 2024) such as Llama (Dubey et al., 2024), DeepSeek (DeepSeek-AI, 2024), Qwen (Qwen Team, 2024), and Grok (Grok, 2024), organizations are integrating LLM inference into their products. However, operating such systems is a high-risk, business-critical task due to substantial costs. As shown in Table 1, serving a single request with Llama3 70B is orders of magnitude more expensive than deploying a typical recommendation model (Naumov et al., 2019), necessitating efficient system designs.

At Meta, Llama empowers nearly one billion monthly active users across various product surfaces (lla, 2025). Through optimizing LLM inference systems for use cases ranging from interactive agents to offline data processing, we accumulated insights that we retrospectively document in this paper. We share *the challenges* of navigating a vast design space, *the exploration methodology* we developed to systematically evaluate design options, and *the core insights* gained throughout the optimization process.

Metrics per query	**Llama3 70B** (2K Input, 150 Output, 64 Batched)	**Recommendation** (Rank 4K Items)
FLOPs	8 EFLOPs+	$<$ 2.5 TFLOPs
Memory traffic	10TB+	$<$ 400GB
System time	Few seconds	$<$ 100 ms

Table 1: Workload characteristics of Llama3 70B and production recommendation models.

The challenge. The complexity of building LLM inference systems arises from the interaction of three factors.

(1) Diverse application requirements: Systems must support a wide range of workload characteristics (input/output lengths, context accumulation, multi-modality) and Service Level Objectives (SLOs), such as Time-To-First-Token (TTFT) and Time-To-Incremental-Token (TTIT) latencies, and throughput. For instance, optimizing for low-latency chatbots requires different configurations than high-throughput offline data scoring.

(2) Combinatorial design space parameters: Deploying LLMs involves a wide range of decisions: selecting hardware accelerators (e.g., NVIDIA H100/H200 (Cho-

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Proceedings of the 9^{th} MLSys Conference (Industry Track), Bellevue, WA, USA, 2026.

quette, 2023; NVIDIA Corporation, 2024b), AMD MI300X (Smith & Alla, 2025)) with different compute/memory profiles and costs, choosing parallelism strategies (Tensor, Pipeline, Expert, Context, Data, or hybrids), designing the runtime and scheduling system (e.g., continuous batching vs. disaggregating prefill and decode phases), and selecting model and system optimization techniques trading off quality and speed. Finding the right combination balancing SLO, cost, and quality is extremely challenging.

(3) Rapid pace of innovation: LLMs evolve rapidly with new model architectures (e.g., MoE, multi-head latent attention (Liu et al., 2024a; DeepSeek-AI et al., 2024)) and optimization techniques. These innovations require re-evaluation of optimal system configurations, making manual or heuristic approaches unsustainable.

The exploration. While specific optimizations (Section 5), simulations (Agrawal et al., 2024a; Cho et al., 2024), and analyses (Yuan et al., 2024) offer valuable insights, systematically exploring the vast configuration space to optimize large-scale LLM inference remains challenging. To address this, we developed a lightweight performance simulator for systematic design space exploration (Section 3). Built on empirical operator benchmarks, our simulator rapidly projects end-to-end performance (latency, throughput) across millions of deployment configurations (hardware, parallelism, runtime). This enables data-driven selection of cost-effective systems tailored to production workloads and SLOs.

The insights. Our experience yielded key insights (Section 4) for optimizing LLM inference system deployments, with throughput improvements up to $\sim 2.5\times$.

Specifically, we discuss 1) runtime design trade-offs, 2) the importance of parallelism strategy, 3) cost-saving opportunities through heterogeneous hardware, 4) system-level implications of new model architectures such as Mixture-of-Experts (MoE), and 5) the impact of hardware infrastructure scaling options.

2 Challenges of Constructing LLM Inference Systems

Designing LLM inference systems requires navigating challenges from the interplay between workload characteristics, infrastructure choices, and optimization options. This section categorizes challenges into workload (Section 2.1), infrastructure (Section 2.2), and optimization (Section 2.3) dimensions.

2.1 Workload Challenges

Production workloads exhibit extreme variability, with each use case dictating different optimization goals based on distinct computational patterns.

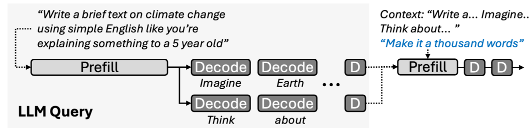


Figure 1: Lifecycle of an LLM query.

2.1.1 *Lifecycle of an LLM Query*

Figure 1 shows the LLM inference lifecycle with *prefill* and *decode* phases. Their different computational characteristics necessitate distinct optimization strategies across hardware, parallelism, and runtime design.

Prefill processes potentially long input sequences, which makes it a compute-bound operation, demanding high compute throughput and dictating TTFT. **Decode** generates tokens one-by-one, making it memory-bandwidth bound due to repeated weight and KV cache access. Low TTIT requires sophisticated batching to amortize memory costs, introducing scheduling trade-offs (Section 4.2). **Context accumulation** in interactive sessions grows KV cache size, increasing costs for both phases, which requires techniques like Persistent KV cache (Gim et al., 2024) to deduplicate computation.

2.1.2 *Attributes Defining LLM Applications*

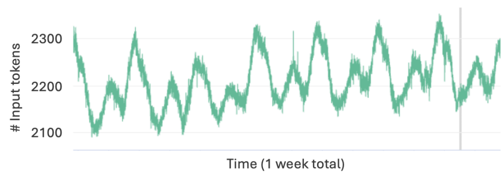
Systematically addressing the diverse landscape of LLM use cases requires defining key workload attributes (Table 2). The variability and interplay of these attributes present significant system design challenges: **Input/output/context length** dictate computational load for prefill and decode. Their extreme variability in production (Figure 2), coupled with hidden costs like system prompts and multi-generation, pose significant challenges for resource allocation and parallelism selection. **Latency SLOs (TTFT/TTIT)** need to be met while minimizing inference costs, which becomes a prohibitively expensive task given different products operate with different SLOs. **Model size** trades quality vs. cost; larger models may deliver higher quality but require complex parallelization and multi-host infrastructure (Section 4.3). **Session-based context accumulation** benefits from persistent KV caching, but introduces significant system complexity involving cache management. Additional factors like multi-modality, real-time patterns, output constraints further differentiate workloads.

2.1.3 *Representative Applications*

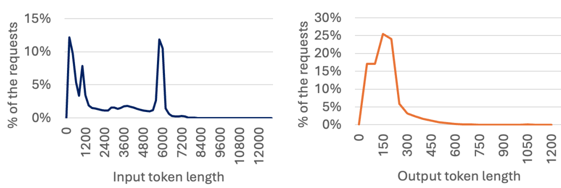
Agents: Interactive agents demand strict TTFT/TTIT SLOs, crucial for user engagement. However, they often exhibit a wide range of input/output lengths. Multi-modal agents add extra complexity via large media inputs, though upload overhead may relax TTFT constraint. Real-time agents (e.g., voice) must meet extremely tight latency SLOs and potentially undefined input boundaries (e.g. continuous voice stream). **Helper models:** Background tasks like content safety checking or query rewriting use smaller models. Input lengths can vary widely due to variety of

↓ Service | Attribute →	Input length	Output length	Context length	TTFT	TTIT	Model size	Multi-Generation	Session-based
Text agent	500-2K	100-500	<128K	200-500ms	<40ms	10-500B	Y	Y
Multi-modal agent	MM/Text 500-2K	100-500	<128K	300-800ms	<40ms	10-500B	Y	Y
Real-time agent	<50	50-100	<32K	<200ms	<30ms	10-100B	N	Y
Safety	<8K	Few	=Input	<200ms	N/A	10B	N	N
Query generation	<8K	100-500	=Input	<200ms	<10ms	10B	N	N
Data augmentation	<8K	<1K	<32K	Few sec	<300ms	100B-500B	Y	Y
Data scoring	<32K	Few	=Input	N/A	N/A	100B-500B	N	N

Table 2: Characterization of diverse production LLM application scenarios and their typical attribute ranges. These variations significantly impact optimal LLM inference system design.

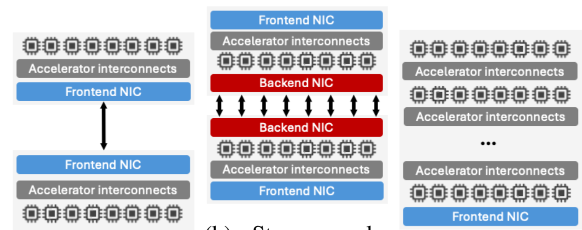


(a) 1-minute average input tokens over time.



(b) Histogram of input and output token length.

Figure 2: Input/Output token lengths from production, showing high variability.



(a) Weak scale-out. (b) Strong scale-out. (c) Scale-up.

Figure 3: Hardware platform architectures with different trade-offs.

applications, but output lengths are typically more constrained and deterministic (e.g., classification, structured generation, etc). Their SLOs are very tight since they sit in the critical path between primary LLMs and external entities. **Offline processing** such as data generation, augmentation, and scoring (Dubey et al., 2024; Xu et al., 2024; Ouyang et al., 2022) are typically latency-tolerant, allowing the use of large, high-quality models along with bigger batches mostly optimized for throughput.

2.1.4 *Production Characterization Challenges*

Real-world LLM workloads exhibit high variability beyond those already laid out in Table 2. Figure 2 illustrates that, while we can consider a text agent to have about ∼2K input tokens based on 1-minute average (Figure 2a), the actual distribution (Figure 2b) spans from brief queries to lengthy documents. This variability, multiplied across all attributes in Section 2.1.2, creates unpredictable patterns, requiring robust systems built with precise performance modeling and systematic exploration (Section 3).

2.2 Infrastructure Challenges

Infrastructure choices span hardware platforms, runtime systems, and parallelization strategies.

2.2.1 *Hardware Platform*

Hardware accelerators differ in compute, memory, and interconnect characteristics. Production datacenters host multiple platform types (Figure 3): **Weak scale-out** (PCIe-connected) suits smaller models cost-effectively. **Strong scale-out** uses high intra-host bandwidth (e.g., NVLink) with moderate inter-host links, balancing scalability and cost. **Scale-up** employs custom high-bandwidth interconnects (e.g., TPU pods (Jouppi et al., 2023), NVLink Switch (Tirumala & Wong, 2024)) for peak performance at higher cost. Wrong platform choice may cause $2-3\times$ cost inefficiency or SLO failures (Section 4.6).

2.2.2 *LLM Inference Runtime Design*

Runtime complexity stems from managing concurrent requests. Figure 4 shows two approaches. *Continuous batching* (Yu et al., 2022; Daniel et al., 2023) executes mixed prefill/decode tasks in unified iterations, maximizing utilization but coupling phases; large prefills can delay decode, violating TTIT SLOs. *Disaggregated approaches* (Zhong et al., 2024; Qin et al., 2024) use separate prefill/decode pools for phase-specific optimization and better SLO predictability (Section 4.2), but add complexity: pool sizing, KV cache transfer, and potential underutilization.

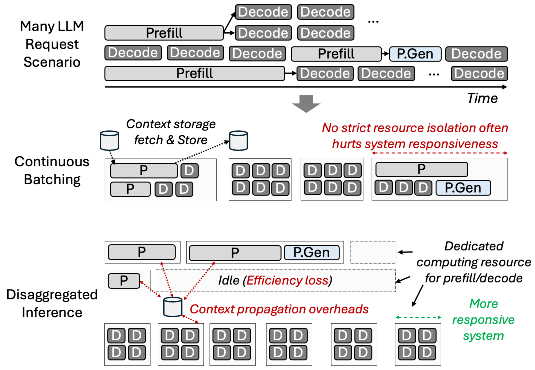
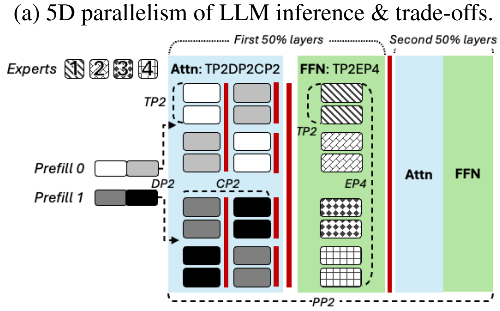


Figure 4: Runtime handling concurrent queries via continuous batching or disaggregation.

Parallelism	Memory capacity		Latency	Throughput	Sync Overhead
	Weight	KV cache			
Tensor	1/N	1/N	1/N	N	High (AllReduce)
Pipeline	1/N	1/N	1	N	Low (Send/Recv)
Expert	1/N	N/A	1/N	N	Med (All2All)
Context	1/N	1/N	1/N	N	Med (Send/Recv)
Data	1	1	1	N	1

(a) 5D parallelism of LLM inference & trade-offs.



(b) Example prefill parallelization scenario, using TP2DP2CP2 (Attention) and TP2EP4 (FFN).

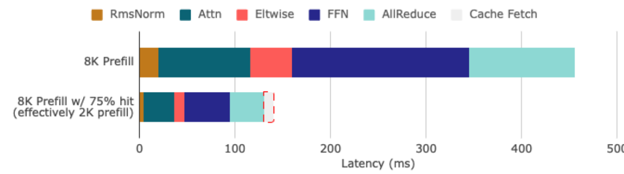
Figure 5: Types of LLM parallelism and an example scenario.

2.2.3 *Parallelization Strategies*

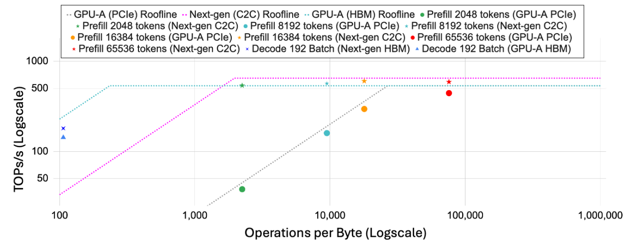
Figure 5 shows five parallelism types with trade-offs: **Tensor Parallelism (TP)** (Shoeybi et al., 2019) shards parameters across devices, improving latency/throughput but requiring high-bandwidth interconnects. **Pipeline Parallelism (PP)** (Lepikhin et al., 2020; Shoeybi et al., 2019) chunks layers sequentially, improving throughput with manageable communication. **Expert Parallelism (EP)** (Lepikhin et al., 2020; Du et al., 2022) distributes MoE experts using All2All, but may suffer load imbalance. **Context Parallelism (CP)** (Liu et al., 2023; Yang et al., 2024) partitions sequences, beneficial for long-context prefill. **Data Parallelism (DP)** (Rajbhandari et al., 2020) replicates weights for independent batch processing when other schemes incur excessive communication.

2.3 Optimization Challenges

Optimization techniques alter performance but introduce trade-offs requiring systematic evaluation.



(a) Persistent KV cache reducing prefill costs.



(b) Rooflines for tiered accelerator memory setups.

Figure 6: Performance impact of LLM optimizations.

2.3.1 *Model-level Optimizations*

Architectural innovations like MLA (Liu et al., 2024a) or Mamba (Gu & Dao, 2023) alter compute profiles, requiring architecture-system co-design iterations. **Model compression** reduces computational/memory demands via pruning, distillation (Blog, 2024), or quantization (FP8/INT8) (Xiao et al., 2023a). Quantization strongly impacts performance-cost tradeoffs by leveraging hardware acceleration and reducing memory bandwidth requirements; emerging 4-bit formats (NVIDIA Corporation, 2024a) promise further gains. Each compression technique trades compression ratio, quality, and hardware efficiency, potentially enabling larger batches or reduced parallelism, necessitating system re-optimization.

2.3.2 *System-level Optimizations*

KV cache management: KV cache memory grows with context length. *Persistent caches* (Gim et al., 2024) reuse stored KV for repeated prompts (Figure 6a), reducing prefill costs but adding memory management complexity. *Paged caches* (Kwon et al., 2023) reduce fragmentation via non-contiguous blocks with page management overhead. *Eviction/compression* (Li et al., 2024b; Zhang et al., 2023; Xiao et al., 2023b; Hooper et al., 2024) trade quality for memory efficiency.

Tiered memory: Modern platforms offer host DRAM (via NVLink C2C (nvg) or PCIe) beyond HBM. This enables hosting much larger models via weight/KV cache offloading techniques. However, its effectiveness depends on achieving sufficient arithmetic intensity. Figure 6b shows NVLink C2C links need ~2K tokens to amortize offloading costs; PCIe needs up to 64K tokens.

Speculative decoding (Leviathan et al., 2023; Miao et al., 2023b) uses draft models for candidate proposals, trading draft accuracy and validation overhead for latency reduction.

Notation	Meaning
$\mathcal{M}$	LLM model as a computational graph.
$\mathcal{D}$	Request distribution specifying input/output lengths, max context length, and session behavior.
$\mathcal{H}$	Hardware setup: Choice of accelerators, platforms, and networks.
$\mathcal{P}$	Parallelization choices (PP, TP, CP, EP, and DP), e.g., `PP2-DP2TP4-TP8` means PP=2, Attention=DP2+TP4, FFN=TP8.
$\mathcal{R}$	Runtime config: continuous batching, prefill-decode disaggregation, max batch size, KV-cache policies (e.g., paged, persistent).
$\mathcal{L}$	Latency SLOs ($\mathcal{L}_{TTFT}$, $\mathcal{L}_{TTIT}$).
$\mathrm{TTFT}(\cdot)$	Estimated TTFT given $\mathcal{M}, \mathcal{D}, \mathcal{H}, \mathcal{P}, \mathcal{R}$.
$\mathrm{TTIT}(\cdot)$	Estimated TTIT given $\mathcal{M}, \mathcal{D}, \mathcal{H}, \mathcal{P}, \mathcal{R}$.
$\mathrm{QPS}(\cdot)$	Estimated throughput, i.e., queries per second (QPS), given $\mathcal{M}, \mathcal{D}, \mathcal{H}, \mathcal{P}, \mathcal{R}$. $\mathrm{QPS}_{\mathrm{cluster}}$ represents normalized throughput for a certain cluster size, e.g., 256 accelerators.

Table 3: Variables used in problem formulation.

3 Exploring Inference System Designs

A systematic approach was essential for addressing the complex, multi-faceted optimization problem described in Section 2. We now describe how we navigated multi-million design choices to identify optimal configurations.

3.1 Formal Problem Formulation

We formalize LLM inference optimization (symbols in Table 3) maximizing throughput subject to latency SLOs as:

$$\begin{aligned} \max_{\mathcal{M},\mathcal{D},\mathcal{H},\mathcal{P},\mathcal{R}} \quad & \mathrm{QPS}_{\mathrm{cluster}}(\mathcal{M},\mathcal{D},\mathcal{H},\mathcal{P},\mathcal{R}) \\ \text{subject to} \quad & \mathrm{TTFT}(\mathcal{M},\mathcal{D},\mathcal{H},\mathcal{P},\mathcal{R}) \le \mathcal{L}_{TTFT} \\ & \mathrm{TTIT}(\mathcal{M},\mathcal{D},\mathcal{H},\mathcal{P},\mathcal{R}) \le \mathcal{L}_{TTIT} \end{aligned}$$

For each service scenario ($\mathcal{D}$), we typically examine several models and hardware platforms ($\mathcal{M}, \mathcal{H}$) with dozens to hundreds of total runtime configurations ($\mathcal{R}$; dozens of batch sizes $\times$ several optimization technique combinations $\times$ prefill and decode), already yielding 1000+ combinations. On top of that, for parallelization ($\mathcal{P}$), we set PP first and explore the rest of 5D parallelism (TP, DP, EP, CP) for Attn and FFN each (Figure 5b) for 128–256 accelerator cards, leading to another $\sim$1000 combinations. The total number of combinations therefore often reaches the order of millions.

3.2 Operator and Block Modeling

Solving the problem formulation above requires accurately estimating the performance metrics for any configuration ($\mathcal{M}, \mathcal{D}, \mathcal{H}, \mathcal{P}, \mathcal{R}$). We developed a lightweight performance simulation framework illustrated in Figure 7, which achieves this via a bottom-up modeling approach.

Micro-benchmarking profiles operators (GEMM, attention, AllReduce, AlltoAll) on hardware $\mathcal{H}$ across input shapes, keeping 100K+ results per hardware.

Operator Performance Model $\mathcal{F}$ builds interpolator/extrapolator $\mathcal{F}(\mathrm{op}, \mathcal{H}, \mathrm{shape})$ from measurements. Multi-dimensional piecewise-linear interpolation provides superior accuracy to analytical models.

Model Assembly instantiates operators with shapes from $\mathcal{P}$ and $\mathcal{D}$, queries $\mathcal{F}$ for latencies, sums along critical paths including communication and runtime overheads, yielding end-to-end latencies for *prefill* and *decode*.

3.3 End-to-End Performance Exploration

Runtime-Specific Metrics calculates SLO metrics based on $\mathcal{R}$. **Continuous batching** couples prefill/decode on shared resources; TTFT depends on prefill, TTIT on mixed prefill/decode latency. **Disaggregated inference** uses independent pools; TTFT/TTIT determined by respective pool performance. $\mathrm{QPS}_{\mathrm{cluster}}$ is derived by finding optimal prefill and decode configurations independently, then computing the accelerator ratio that balances their processing rates.

Search Space Pruning. To handle millions of configurations, we prune early: configurations violating hardware constraints (memory capacity) or SLO targets are discarded, and we focus on power-of-two parallelism degrees matching standard 8-GPU topologies. The framework supports arbitrary counts for non-standard topologies (e.g., GB200 NVL72).

Systematic Search iterates through the pruned ($\mathcal{H}, \mathcal{P}, \mathcal{R}$) combinations, generating a "performance landscape" mapping configurations to TTFT, TTIT, $\mathrm{QPS}_{\mathrm{cluster}}$.

Dynamic and Extended Modeling. Beyond static configurations, the framework models speculative decoding (scaling batch sizes by draft tokens and acceptance rates; incorporating empirically observed acceptance rates) and MoE expert routing (using empirical token distributions to capture load imbalance, see Section 4.5). Energy optimization is supported by modeling power-capped accelerators as distinct hardware types; TCO incorporates hardware costs, power, and operational factors.

SLO-Aware Ranking: Configurations violating the latency SLOs ($\mathcal{L}_{TTFT}$, $\mathcal{L}_{TTIT}$) are filtered out. The remaining valid configurations are ranked based on the optimization objective (e.g., highest $\mathrm{QPS}_{\mathrm{cluster}}$). This allows identification of Pareto-optimal frontiers and the corresponding best configuration under given constraints (e.g., hardware availability and cost budgets).

3.4 Evaluation of the Simulator

Accuracy: Figure 8 compares estimated latency against real hardware measurements. The estimated latencies across diverse operators and parallelism setups align closely with measured values, typically within a $\pm$5% error margin. Such high accuracy stems from the simulator's benchmark-driven nature; the majority of performance estimations are minor interpolations of real hardware measurement results. While non-deterministic factors like network jitter can affect P99 tail latencies, our approach reli-

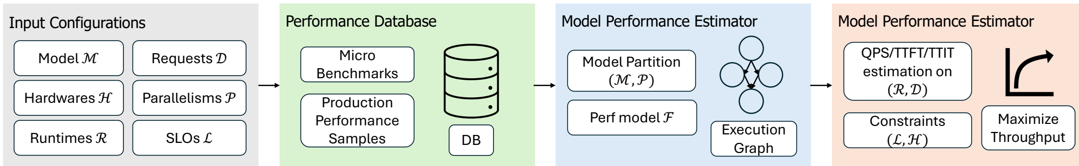


Figure 7: Lightweight LLM inference performance simulator.

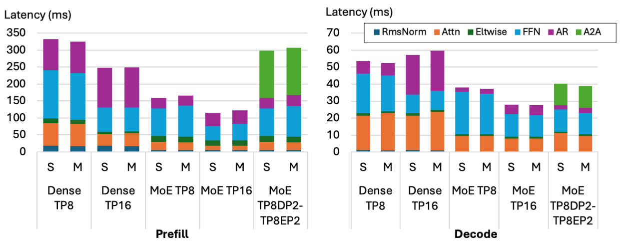


Figure 8: Comparison of simulation (S) and measurement (M) results for various parallelism scenarios.

ably predicts median and mean performance for production deployment decisions.

Speed: Our lightweight benchmark-powered simulator is extremely fast in exploring the design space, leveraging caches and multiprocesses to reduce runtime to several minutes, which is much faster than cycle-level, analytical, or ML-based simulators (Yuan et al., 2024; Agrawal et al., 2024a; Qi et al., 2017).

Emulating Future Hardware: For future hardware that we cannot benchmark, we upload simulation-based operator performance estimations to our database. This provides decent fidelity given LLMs have limited operator types (typically dozens) and minimal caching effects between operators.

4 Key Insights from Case Studies

We now present the five key insights previewed in Section 1, which we derived from our experience and believe provide guidance for designing efficient LLM inference systems.

4.1 Experimental Setup

Target hardware is detailed in Table 4, covering scale-out platforms (Figure 3b) and next-gen scale-up platforms (Figure 3c). The target models and inference scenarios are described in Table 5. Our discussions reflect the models and setups we worked with during this deployment period. All results are from our simulation framework (Section 3). We evaluate configurations meeting latency SLOs ($\mathcal{L}_{TTFT}$, $\mathcal{L}_{TTIT}$) while maximizing throughput ($\text{QPS}_{\text{cluster}}$, normalized per 256 accelerators).

Hardware	TF/s	TB/s	GB	Topology (intra/inter-host bandwidth)
GPU-A	1000	2.4	90	Scale-out (450GB/s, 50GB/s)
GPU-B	1000	4.8	141	Scale-out (450GB/s, 50GB/s)
GPU-C	850	5.2	192	Scale-out (448GB/s, 50GB/s)
NextGen	1800	7.7	189	Scale-out (900GB/s, 50GB/s) Scale-up (900GB/s, 100GB/s)

Table 4: Hardware Platforms ($\mathcal{H}$) used in the case studies.

Variable	Values Explored
$\mathcal{M}$	Llama3 (70B, 405B), MoE variants (405B, 1.8T hypothetical). All assumed FP8 precision for FFN weights unless stated.
$\mathcal{D}$	Varied per scenario (Online: 2K input, 150 output, 8K context; Offline: 2K input, 1K output, 8K context; etc.).
$\mathcal{H}$	GPU-A, GPU-B, GPU-C (Current Gen Scale-out, 8 cards/host), Next-Gen (Scale-out 8 cards/host, Scale-up 64 cards/pod). Projections for Next-Gen.
$\mathcal{P}$	Up to 256 cards. Powers-of-2 parallelism degrees. Combinations of PP, TP, DP (for Attention/FFN), EP (for FFN).
$\mathcal{R}$	**Run Types:** Continuous batching (`cont.batch`) or Disaggregated inference (`disagg`). **Batch Sizes:** Prefill=1 (typical for online); Decode=1 to 4096. **KV cache:** Persistent KV for session-based workloads.

Table 5: LLM inference scopes for the case studies.

4.2 Runtime Design Trade-offs (Latency vs. QPS)

Insight: Disaggregated runtimes excel for online scenarios with strict latency constraints, while continuous batching suffices for offline throughput-oriented workloads.

Impact: Meta migrated the majority of online services to disaggregated runtime to gain around 30% capacity savings, while still leveraging continuous batching for offline services.

The choice of runtime architecture ($\mathcal{R}$), specifically continuous batching (`cont.batch`) versus disaggregated inference (`disagg`), significantly impacts performance depending on the target SLOs ($\mathcal{L}$). To quantify these trade-offs, we used our simulator (Section 3) to identify the optimal configurations for online inference targeting representative strict latency SLOs ($\mathcal{L}_{TTFT}$, $\mathcal{L}_{TTIT}$) for both Llama3 70B and 405B models across various hardware platforms ($\mathcal{H}$). Table 6 summarizes these results, showing the best achievable throughput ($\text{QPS}_{\text{cluster}}$) for both disaggregated (`disagg`) and continuous batching (`cont.batch`) runtimes ($\mathcal{R}$) while meeting the latency constraints.

As shown in Table 6, for online inference with strict latency targets, `disagg` consistently achieves higher throughput while meeting SLOs. For 70B, `disagg` is 1.5-1.8x higher

Hw	Runtype	Setup	Dec. BS	TTFT / TTIT	Prefill / Decode QPS	QPS
		70B Llama3				
GPU-A	Disagg Cont.Batch	Prefill PP4-TP2 Decode TP8 TP8	112 28	205/29 95/30	625/826 -	356 198
GPU-B	Disagg Cont.Batch	Prefill PP4-TP2 Decode TP8 PP2-TP4	128 16	205/28 134/27	625/974 -	381 246
GPU-C	Disagg Cont.Batch	Prefill PP4-TP2 Decode TP8 PP2-TP4	128 16	233/29 143/27	549/927 -	345 248
Next-Gen	Disagg Cont.Batch	Prefill PP8 Decode TP8 PP2-TP4	224 32	214/29 86/29	1198/1660 -	696 469
		405B Llama3				
GPU-A	Disagg Cont.Batch	Prefill TP8 Decode TP8 TP8	16 4	365/44 367/42	88/77 -	41 19
GPU-B	Disagg Cont.Batch	Prefill TP8 Decode TP8 TP8	56 8	365/43 369/43	88/276 -	67 37
GPU-C	Disagg Cont.Batch	Prefill TP8 Decode TP8 TP8	56 8	375/43 380/41	85/277 -	65 39
Next-Gen	Disagg Cont.Batch	Prefill PP2-TP4 Decode TP8 TP8	128 16	349/43 226/42	183/638 -	142 78

Table 6: Optimal *online inference* setups for Llama3 70B and 405B under strict latency targets. Each QPS denotes $\mathrm{QPS}_{\mathrm{cluster}}$.

than `cont.batch`; for 405B, the gap widens to 1.8-2.2x. This stems from two factors: (1) independent optimization of prefill/decode resources and parallelism ($\mathcal{P}$), avoiding `cont.batch` compromises (Section 4.3); (2) significantly larger decode batch sizes (e.g., 112 vs 28 for 70B on GPU-A; 56 vs 8 for 405B on GPU-B) without impacting TTIT.

Conversely, for offline generation (Table 7), where throughput is the sole objective, the gap diminishes. Both converge to deep PP with very large batches. For 70B on GPU-C, `cont.batch` even slightly outperforms `disagg`. Given `disagg`'s operational complexity, continuous batching often becomes more practical for latency-insensitive tasks.

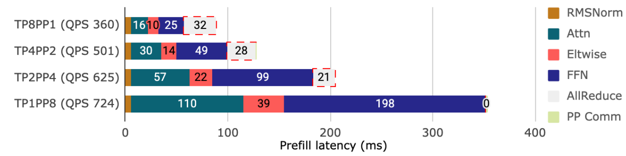
4.3 Actively Leveraging Parallelism Strategy

Insight: Parallelism has a huge impact on overall service throughput due to nontrivial communication overheads. If SLO is easy to meet, setups incurring lower communication overhead help recoup throughput. Disaggregated runtime can further leverage distinct parallelism setups for prefill and decode.

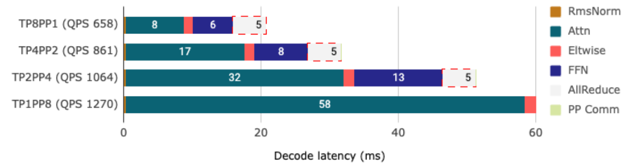
Impact: Meta actively adopted per-service-tuned parallelization setups to maximize the throughput at given SLO.

Optimal parallelism ($\mathcal{P}$) must be tailored to prefill/decode computational demands and target SLOs.

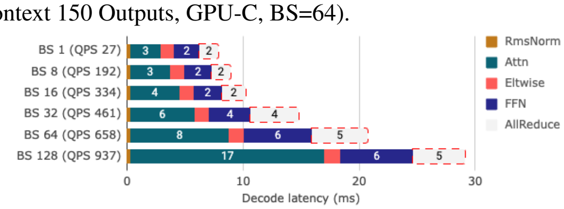
Prefill Parallelism: As shown in Figure 9a for online prefill (70B, 2K tokens, GPU-A), increasing TP degree reduces latency. However, the throughput gains ($\mathrm{QPS}_{\mathrm{cluster}}$) are sub-linear because operators tend to scale sub-linearly and also because of higher communication overheads (i.e. AllReduce). For example, TP4PP2 achieves lower latency than TP2PP4 but yields 20% lower QPS. Therefore, the optimal prefill strategy, as seen in the disaggregated setups in Table 6 (e.g., PP4-TP2 for 70B on GPU-A), aims to efficiently utilize the available $\mathcal{L}_{TTFT}$ headroom, rather than minimizing latency at any cost. Different hardware or workloads might favor different strategies (e.g., Context



(a) Prefill performance for various parallelism setups (70B, 2K Tokens, GPU-A).



(b) Decode performance for various parallelism setups (70B, 8K Context 150 Outputs, GPU-C, BS=64).



(c) Decode performance for various batch sizes (70B, 8K Context 150 Outputs, GPU-C, TP=8).

Figure 9: Online Inference Latency. $\mathrm{QPS}_{\mathrm{cluster}}$ is shown.

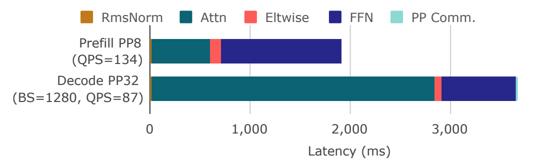


Figure 10: Offline inference Prefill and Decode performance (405B on GPU-A, at ideal setup, $\mathrm{QPS}_{\mathrm{cluster}}$).

Parallelism (CP) becomes relevant for extremely long sequences, not shown here).

Decode Parallelism: For online decode (70B, 8K context, GPU-C), performance hinges on both parallelism ($\mathcal{P}$) and batch size ($\mathcal{R}$). Figure 9b shows that increasing TP (e.g., TP8 vs. TP4) improves latency and throughput by leveraging more aggregate memory bandwidth, crucial for the memory-bound decode phase. Figure 9c demonstrates that increasing batch size boosts QPS initially, but with diminishing returns as computation becomes relatively more dominant. The optimal setup identified (e.g., TP8 with Batch Size 128 for 70B on GPU-C in Table 6) maximizes the throughput within the $\mathcal{L}_{TTIT}$ constraint. Note that the optimal point found often differs from the optimal prefill strategy (e.g., PP4-TP2) or what might have been chosen heuristically, stressing the importance of our systematic approach.

Phase-Specific Optimization: The ability of disaggregated runtimes to employ distinct parallelism strategies for prefill and decode (e.g., PP4-TP2 for prefill, TP8 for decode in the 70B GPU-A online case from Table 6) is a primary reason for their superior efficiency in

Hw	Runtype	Setup	Dec. BS	TTFT / TTIT	Prefill QPS/ Decode QPS	QPS
		70B Llama3				
GPU-A	Disagg Cont.Batch	Prefill PP8 Decode PP32 PP16	896 512	353/1254 1071/894	724/179 -	143 143
GPU-B	Disagg Cont.Batch	Prefill PP8 Decode PP8 PP8	256 256	353/323 711/420	724/198 -	155 152
GPU-C	Disagg Cont.Batch	Prefill PP8 Decode PP64 PP16	2560 1024	407/2300 1301/1300	629/278 -	193 197
Next-Gen	Disagg Cont.Batch	Prefill PP8 Decode PP8 PP8	256 256	214/211 436/267	1198/303 -	242 239
		405B Llama3				
GPU-A	Disagg Cont.Batch	Prefill PP8 Decode PP64 PP32	1280 640	1918/3675 3777/3050	134/87 -	53 52
GPU-B	Disagg Cont.Batch	Prefill PP8 Decode PP16 PP16	256 256	1918/643 2660/1145	134/100 -	57 56
GPU-C	Disagg Cont.Batch	Prefill PP8 Decode PP128 PP32	4096 1024	2008/8420 4060/4058	127/122 -	62 63
Next-Gen	Disagg Cont.Batch	Prefill PP8 Decode PP8 PP8	256 256	1134/418 1588/709	226/153 -	91 90

Table 7: Optimal *offline inference (generation)* setups for Llama3 70B and 405B. Each QPS denotes $\mathrm{QPS}_{\mathrm{cluster}}$.

latency-constrained online scenarios compared to continuous batching. For offline tasks (Table 7), where latency is not critical, the optimal strategy often converges to deep PP for both phases, combined with very large batch sizes to maximize throughput (Figure 10).

Interaction with Hardware Topology and Capability: The effectiveness of parallelism strategies is also modulated by the hardware topology and capability. TP demands strong interconnects (e.g., intra-host NVLink or scale-up fabrics), whereas PP and EP (Expert Parallelism, see Section 4.5) are more tolerant of weaker inter-host links. The models in Table 6 fit within a single host (8 GPUs) capacity-wise and also seem to prefer not going beyond a single host to avoid costly inter-host communications. Larger models going beyond a single host's capacity *necessitate* multi-host deployments which we explore in Section 4.6 and Section 4.5. The impact of hardware capabilities on the optimal strategy is also evident in Figure 11, where the Next-Gen platform with faster speed meets SLO with less aggressive prefill (PP2-TP4) and decode (TP8 with larger batch size) setups, compared to GPU-A which is forced to use TP8 to meet SLO.

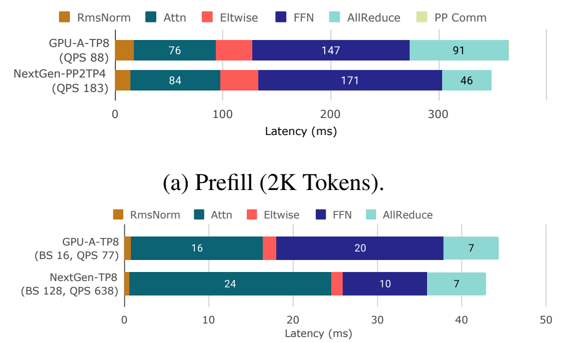
4.4 Hardware Heterogeneity Opportunities

Insight: For disaggregated systems, strategically mapping compute (prefill) and memory bandwidth (decode) intensive phases to accelerators of different strengths can improve the overall cost efficiency by 15-25% compared to homogeneous deployments.

Impact: Meta deployed services to platforms that best fit the characteristics to maximize the accelerator fleet efficiency.

Prefill tends to be compute-bound, favoring high FLOP/s accelerators (GPU-A). Decode is memory-bandwidth bound, benefiting from high HBM bandwidth (GPU-B/C). Homogeneous clusters may be suboptimal as no single type suits both phases. Disaggregated runtimes (Section 4.2) enable leveraging heterogeneity.

Table 8 explores heterogeneous configurations for the 405B online inference scenario using disaggregation. First, homogeneous setups show good potential; GPU-A excels at prefill (88 QPS) but struggles with decode (77 QPS); GPU-B/C are weaker at prefill (88/85 QPS) but much stronger at decode (276/277 QPS). In our production deployments, the heterogeneity may come from either different vendors or different products from the identical vendor. Heterogeneous configurations, specifically using GPU-A for prefill and GPU-B or GPU-C for decode, achieve end-to-end QPS ($\mathrm{QPS}_{\mathrm{cluster}}$ = 67) comparable to the best homogeneous setups (GPU-B/B or GPU-C/C). This implies potential cost savings – if the combined cost of GPU-A (for prefill) and GPU-B/C (for decode) resources needed to achieve 67 QPS is lower than the cost of GPU-B/C resources alone for the same throughput, heterogeneity wins. Based on the performance characteristics and actual cost structures, our performance modeling estimates suggest potential TCO improvements of 15-25%. Note that the optimal ratio of prefill-to-decode accelerators also varies significantly depending on the hardware combination (e.g., 0.88 for A/A vs. 3.14 for A/B).



(a) Prefill (2K Tokens).

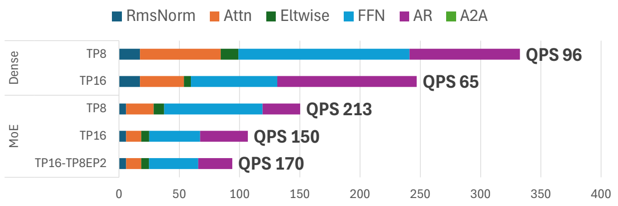
(b) Decode (8K Context, 150 Output)

Figure 11: 405B online performance comparison for GPU-A and Next-gen at their optimal setups. $\mathrm{QPS}_{\mathrm{cluster}}$ is shown.

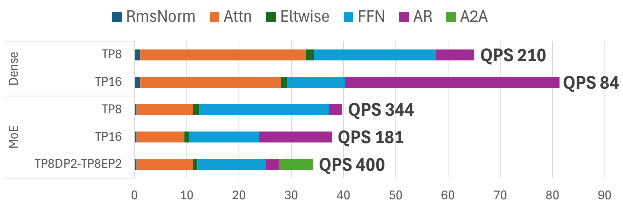
Prefill HW	Decode HW	Prefill QPS	Decode QPS	Prefill:Decode	End-to-end QPS
GPU-A	GPU-A	88	77	0.88	41
GPU-B	GPU-B	88	276	3.14	67
GPU-C	GPU-C	85	277	3.26	65
GPU-A	GPU-B	88	276	3.14	67
GPU-A	GPU-C	88	277	3.16	67

Table 8: Heterogeneous inference for 405B model. Each QPS denotes $\mathrm{QPS}_{\mathrm{cluster}}$.

Practical Considerations: While promising for cost efficiency, implementing heterogeneous inference introduces operational complexities in capacity planning, scheduling across distinct hardware pools among geographically distributed datacenters of Meta. However, exploiting these differences via disaggregation remains valuable, as datacenters naturally incorporate diverse hardware generations.



(a) Prefill performance breakdown (2K tokens).



(b) Decode performance breakdown (BS=64, 8K context).

Figure 12: Performance of 405B Dense and iso-sized MoE (125B x 4 Experts, Top 2 triggered per token) on GPU-A. $\mathrm{QPS}_{\mathrm{cluster}}$ is shown.

4.5 MoE vs. Dense Performance and Parallelism

Insight: Mixture-of-Experts (MoE) models achieve higher computational efficiency than dense counterparts of equivalent total parameter counts by activating only a subset of parameters per token. However, unlocking this potential, particularly when scaling across multiple hosts, hinges critically on adopting appropriate parallelism strategies, primarily Expert Parallelism (EP) over Tensor Parallelism (TP) for FFN layers, especially on scale-out infrastructure.

Impact: This exploration added great confidence for Meta to migrate the majority of models to MoE-based architectures. We are also leveraging the parallelism related lessons in our fleet.

Computational Efficiency: The fundamental advantage of MoE models stems from their sparse activation pattern. By processing each token through only a subset of FFN "experts", MoE models significantly reduce the per-token FLOP count compared to dense models where every token passes through the entire FFN layer. This benefit is most pronounced in compute-bound phases like prefill.

We compare a 405B dense Llama3 model with a hypothetical iso-parameter MoE variant (125B base x 4 experts, activating top 2 experts per token) on GPU-A. We advise to compare MoE and dense with *the identical parallelism setup* for fair assessment. Note that we assume this hypothetical MoE model architecture mainly for performance/efficiency study; it does not represent a sound/effective expansion of the Llama3 405B model.

As shown in Figure 12a, the MoE model achieves significantly lower prefill latency and higher throughput than the dense model using the same TP8 parallelism (i.e., QPS 96 vs. 213) within a single 8-card host, directly reflecting the reduced FFN computation.

For the memory-bandwidth-bound decode phase (Figure 12b), the advantage of MoE should diminish compared to prefill. Specifically, for MoE model with large enough decode batch size far exceeding the number of experts, tokens will likely trigger all experts, necessitating accesses to the full set of expert weights, thus eliminating the benefits of sparse FFN activation (vs. Dense). However, in our specific comparison using an iso-parameter MoE construction, we still observe a notable decode speedup; for example, TP8 MoE achieves much better latency and QPS vs. Dense counterpart (i.e., 344 vs. 210). This is largely attributable to the smaller *non-expert* parameters, particularly the attention layers, in our hypothetical MoE model compared to its dense counterpart, which significantly reduces the memory footprint and computation required for the attention part of the model.

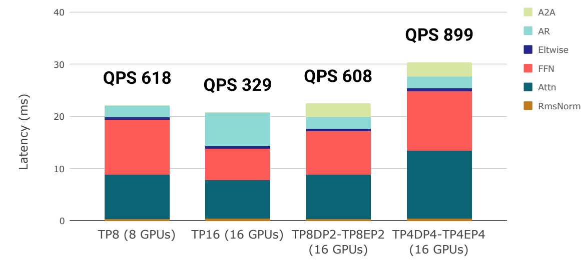


Figure 13: Llama 4 Maverick decode performance under different parallelism strategies on GPU-A (BS=64, 8K context). EP-based scaling preserves throughput across hosts, unlike pure TP which suffers from AllReduce overhead. $\mathrm{QPS}_{\mathrm{cluster}}$ is shown.

Parallelism Strategy and MoE: Realizing MoE's benefits at scale, especially across multiple hosts, necessitates choosing the right parallelism strategy ($\mathcal{P}$). While TP is effective for dense models within tightly-connected nodes (e.g., NVLink within a host), its reliance on costly AllReduce collectives makes it scale poorly across standard inter-host networks (Section 4.6). MoE architectures enable *Expert Parallelism (EP)*, where different experts are placed on different devices/hosts. EP typically requires AlltoAll communication to route tokens to their assigned experts and gather results, which is often significantly more efficient than AllReduce over weaker interconnects.

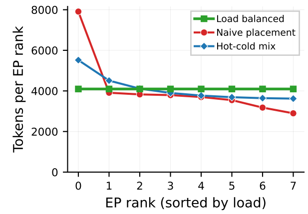
Figure 12b (Decode) best illustrates the benefit of EP. For Dense model, TP8 to TP16 actually degrades both the QPS (210→84) and the latency due to big AllReduce overheads. Similar behavior is observed for MoE model trying to utilize TP16. In contrast, with EP, MoE model improves both the throughput (344→400) and the latency thanks to EP's lower communication costs. We validate this on Llama 4 Maverick (128 routed experts, top-1), part of a growing trend toward many-expert MoE architectures (e.g., DeepSeek-V3 (DeepSeek-AI et al., 2024) with 256 experts) where EP becomes essential (Figure 13). Scaling from TP8 (single host) to TP16 (two hosts) nearly halves decode $\mathrm{QPS}_{\mathrm{cluster}}$ (618→329) because AllReduce costs triple. In contrast, TP8DP2-TP8EP2 preserves throughput (608, −2%) by replacing inter-host FFN AllReduce with cheaper All2All for expert routing. Going further, TP4DP4-TP4EP4 actually achieves the highest throughput (899, +45% over TP8 baseline) by minimizing the communication overheads while maximizing the number of tokens being handled per iteration (i.e., large DP).

For prefill (Figure 12a), TP16-TP8EP2 benefits MoE via reduced FFN AllReduce costs. However, multi-host doesn't improve compute-bound prefill throughput due to communication overhead. Memory-bound tasks benefit from multi-host as token count scales with hosts and weight-reading distributes across devices.

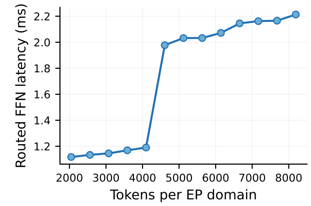
System Implications: When deployed with appropriate parallelism strategies, MoE models offer a compelling path towards higher performance-per-cost, especially on standard scale-out infrastructure with relatively weak inter-host bandwidths. Our findings hold across both our controlled iso-parameter study (Figure 12) and production architectures like Llama 4 Maverick (Figure 13): EP-based parallelism consistently outperforms pure TP for cross-host MoE scaling. Thorough inference space exploration (Section 3) is essential for identifying the optimal parallelism setup (e.g., TP within a host combined with EP/DP across hosts) and striking the right balance of efficiency and latency.

Expert Load Imbalance: MoE performance modeling requires accurately estimating how many tokens each expert gets. For prefill, with many tokens routed across experts, most or all experts are typically activated. The key concern is the *distribution* of tokens among experts: in practice, a few "hot" experts receive disproportionately many tokens (e.g., in one representative Llama 4 Maverick workload, the busiest expert received ∼7% of all routed tokens out of 128 experts). Since per-EP-domain routed FFN latency grows with token load (Figure 14b), the most-loaded EP domain becomes the latency bottleneck. This can be mitigated by mixing hot and cold experts within the same EP domain (Figure 14a), bringing per-domain load closer to the *load-balanced* ideal, i.e., an even split of tokens across experts. Our framework models prefill using this load-balanced assumption, which provides a lower bound on cost.

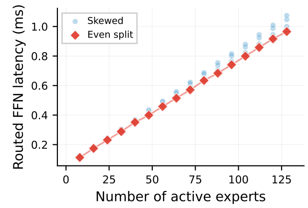
For decode, the dominant cost driver is different. Because decode processes few tokens per expert, routed FFN latency is memory-bandwidth bound and dominated by expert weight reads, scaling linearly with the number of active experts (Figure 14c). Imbalance has a dual effect: when not all experts are triggered, skewed routing leaves some experts inactive, *reducing* cost by avoiding weight reads. Once most experts are active, uneven token counts *within* them add modest overhead (scatter points above the balanced line in Figure 14c). To predict the number of active experts, our framework supports three activation models (Figure 14d): *load balanced* (as in prefill), *uniform random* (each token independently picks an expert), and *log-linear* (calibrated from production traffic). As shown in Figure 14d, the models diverge significantly at moderate batch sizes, leading to up to 3.5× difference in predicted cost; we default to log-linear, which best matches production traffic.



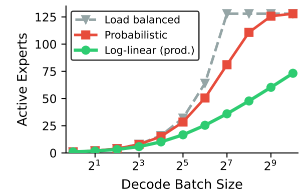
(a) EP-rank load under different placements.



(b) Routed FFN latency vs. EP-domain load.



(c) Decode routed FFN cost vs. active experts.



(d) Expert activation models (decode).

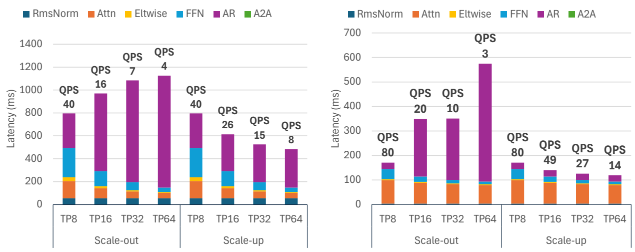
Figure 14: Expert load imbalance sensitivity (Llama 4 Maverick, 128 experts, GPU-A). **Prefill (top):** (a) Per-EP-rank token load under three expert placement strategies (T=32K, EP8, 16 experts/domain). (b) Routed expert FFN latency vs. EP-domain token load (EP8). **Decode (bottom):** (c) Routed expert FFN latency vs. active expert count, with balanced and imbalanced token distributions among active experts (TP8, T=256). (d) Predicted active expert count vs. decode batch size under three activation models.

4.6 Scale-out vs. Scale-up platform tradeoffs

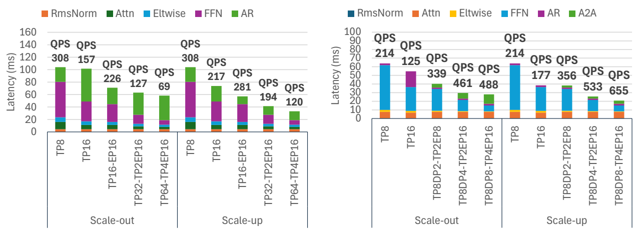
Insight: Scale-out platforms offer cost efficiency for models serviceable within a single or few hosts. Decode of MoE models scales particularly well on scale-out platforms too. Scale-up platforms become necessary for extremely large dense models or extra low latency inference of MoE or Dense models. Their order-of-magnitude higher bandwidth and lower latency vs. scale-out facilitates enabling wider parallelism domain sizes with tolerable communication overheads.

Impact: Meta proactively assessed the potential deployment scenarios of scale-up platforms and the corresponding benefits/challenges using the lightweight performance simulation and design space exploration methodology.

We evaluate whether expensive scale-up infrastructure (high-bandwidth, low-latency interconnects within large accelerator pods) justifies its cost versus conventional scale-out systems (smaller hosts with standard network fabrics) for extremely large LLMs. Using our simulator, we evaluated hypothetical 1.8T Dense and MoE models on Next-Gen platforms: scale-out (8 cards/host, 50 GB/s inter-



(a) Dense model Prefill (Left) and Decode (Right).



(b) MoE model Prefill (Left) and Decode (Right)

Figure 15: 1.8T dense/MoE model performance on scale-out and scale-up next-gen platforms. $\mathrm{QPS}_{\mathrm{cluster}}$ is shown.

host) and scale-up (64 cards/pod, 900 GB/s intra-pod). Results are shown in Figure 15.

Dense Model Scaling Challenges: Large dense models require high TP degrees, relying on expensive AllReduce. Figure 15a shows scale-out TP scaling severely hampered beyond single host due to AllReduce costs, yielding no latency/throughput gains. Scale-up handles multi-node TP effectively with high-speed fabric, although returns diminish at high TP degrees (32-64 cards) as communication dominates.

MoE Model Scaling Advantages: MoE models offer an alternative scaling path via Expert Parallelism (EP), distributing FFN experts across devices and primarily relying on AlltoAll communication, often less demanding than TP's AllReduce. Figure 15b shows MoE scales gracefully with EP on scale-out, especially for Decode (Section 4.5). Prefill throughput scaling remains limited. Multi-host scale-out becomes viable for MoE unlike dense models suffering from communication overheads.
Scale-up yields superior MoE performance via faster AlltoAll and hybrid parallelism (EP across nodes, TP within), achieving the best latency and throughput at large scales.

System Implications: Optimal setup choice involves complex tradeoffs between model size, architecture (Dense vs. MoE), SLOs, parallelism, and platform TCO. Scale-up is necessary for latency-critical large dense models or extremely low-latency MoE; scale-out remains viable for MoE, avoiding high TCO and larger failure domains.

5 Discussion and Related Work

Our work builds on prior research in LLM inference systems, parallelism, and performance modeling.

Basic Inference Systems and Runtime Concepts. Prior work includes general optimizations (NVIDIA, 2017; Aminabadi et al., 2022), continuous batching (Yu et al., 2022), KV cache management (Kwon et al., 2023; Lin et al., 2024; Liu et al., 2024b; Hooper et al., 2024), and specialized runtimes (ModelTC, 2024; Vaidya et al., 2023). Distinct prefill/decode profiles motivated disaggregated architectures (Zhong et al., 2024; Patel et al., 2024; Qin et al., 2024; Agrawal et al., 2023; 2024b) and offloading (Sheng et al., 2023; Jiang et al., 2024). Our methodology systematically evaluates these choices (`cont.batch` vs. `disagg`), quantifying trade-offs for specific workloads, hardware, and SLOs (Sections 4.2 and 4.4), validating disaggregation's benefits for latency-critical tasks.

Advanced Serving Systems. Open-source systems like vLLM (Kwon et al., 2023) and TensorRT-LLM (Vaidya et al., 2023) continue to provide optimizations on kernels, memory management, and scheduling for LLM deployment. Recent work includes Sarathi-Serve (Agrawal et al., 2024b) (chunked-prefills) and Niyama (Kamahori et al., 2025) (disaggregated scheduling). Our work complements these: while they optimize *how* to execute inference, we address *what* configuration to deploy—parallelism strategies, hardware selection, and runtime architecture choices that apply across serving systems.

Parallelism Strategies. Scaling LLMs leverages TP, PP (Shoeybi et al., 2019; Narayanan et al., 2021), EP for MoE (Lepikhin et al., 2020; Shazeer et al., 2016; Fedus et al., 2022; Liu et al., 2024a; DeepSeek-AI et al., 2024; Rajbhandari et al., 2022; Hwang et al., 2023), CP for long contexts (Li et al., 2021; Korthikanti et al., 2023; Jacobs et al., 2023; Liu et al., 2023; Brandon et al., 2023; Yang et al., 2024), and DP (Rajbhandari et al., 2020; Ren et al., 2021; Zhao et al., 2023). Our contribution (Sections 4.3, 4.5 and 4.6) empirically demonstrates that optimal parallelism depends critically on inference phase, model architecture (Dense vs. MoE), and hardware platform (scale-out vs. scale-up). We showed MoE models with EP scale effectively on scale-out systems and identified optimal hybrid strategies.

Performance Modeling and Optimization. Performance prediction via analytical models (Yuan et al., 2024; Bambhaniya et al., 2024), simulators (Agrawal et al., 2024a), or performance models (Qi et al., 2017) enables design space exploration alongside optimizations like quantization (Li et al., 2024a; Xiao et al., 2023a), pruning (Frantar & Alistarh, 2023; Ma et al., 2024), and speculative decoding (Leviathan et al., 2023; Miao et al., 2023b; Xia et al., 2024) (surveys: (Zhou et al., 2024; Miao et al., 2023a; Wan et al., 2023)). Our lightweight simulator (Section 3) integrates empirically-grounded operator models with system-level configurations (runtime, parallelism) for rapid, comprehensive exploration, providing practical guidance (Section 4).

Applicability of Insights on Other Inference Scenarios. The general tradeoffs we identify—TP vs. PP for different phases, EP benefits for MoE on scale-out, disaggregation for SLO-sensitive workloads—stem from fundamental compute and communication characteristics. The magnitude varies with context: parallelism matters less for smaller models on fewer accelerators, while latency-sensitive applications require optimizations that are optional for throughput-focused workloads. Our insights provide directional guidance; practitioners should validate configurations in their specific environments.

Future Directions. The drive for efficiency fuels innovations toward novel architectures (Gu & Dao, 2023; Peng et al., 2023) and optimizations (context reuse, sparsity), with prefill/decode management (Section 2.1.1) remaining central. The compute (prefill) vs. memory bandwidth (decode) tension may favor specialized hardware, though fungibility concerns persist. Future accelerators (Firoozshahian et al., 2023; Jouppi et al., 2023) with balanced performance and fine-grained partitioning could enable better co-scheduling, improving upon current disaggregation approaches.

6 Conclusion

This paper documented the real-world challenges we encountered optimizing LLM inference systems at scale. We systematically defined the optimization space and developed an effective design space exploration methodology based on lightweight simulations, yielding five critical insights that helped Meta optimize production LLM inference services. We hope our lessons—encompassing online and offline inference, dense and MoE models, and various accelerator platforms with different capabilities—help others construct efficient LLM serving systems.

Acknowledgments

We would like to acknowledge all the contributors who made this work possible. Special thanks to Mahesh Balasubramanian for thoroughly reviewing the manuscript and providing valuable feedback.

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